Exposé on Master's Thesis

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1 Topic

Autonomous vehicles are rapidly becoming a reality. However, their perception systems still face significant challenges in complex and unforeseen scenarios, often leading to disengagements, where human intervention via teleoperation is required. In such critical moments, the vehicle must rely on an accurate and robust perception system to ensure safe navigation.

Most existing perception systems, responsible for detecting and tracking objects in the vehicle's surroundings, rely heavily on learning-based solutions. While these methods have demonstrated impressive performance in structured and controlled environments, they often struggle to generalize in novel and untrained scenarios. This limitation is particularly problematic in safety-critical situations, where the vehicle must make quick decisions based on the perceived environment.

This project aims to develop a deterministic, stereo camera-based perception system for autonomous vehicles, capable of providing a reliable and robust understanding of the vehicle's surroundings, even in highly dynamic and unforeseen scenarios. The system will take as input two images from a stereo camera mounted on the vehicle and output a list of detected objects in the scene, along with their criticality scores, forming a situational awareness module to assess potential hazards. Additionaly, this deterministic approach could run in parallel with existing learning-based perception systems, providing a an additional layer of redundancy and security in critical situations where the learning-based methods may fail.

To achieve accurate 6D motion estimation (position + velocity), the system will implement a deterministic algorithm inspired by the Daimler 6D Vision system. This approach uses Kalman Filters to estimate a 6D state vector for selected feature points in the scene, which will then be clustered to identify moving objects. Unlike learning-based methods, this approach does not require extensive training data and can generalize well in new unseen scenarios.

Given the real-time requirements of autonomous vehicles, the system will be implemented in C++ using the Robot Operating System (ROS) framework. It will also incorporate GPU hardware acceleration using CUDA to speed up computationally intensive tasks, such as feature point detection, clustering or optical flow computation.

To ensure scalability and modularity, the entire system will be containerized using Docker, facilitating deployment across different hardware configurations. Finally, the developed system will be deployed and evaluated on the EDGAR research vehicle. Here, the system's performance will be compared against the state-of-the-art learning-based perception systems currently used in the vehicle.

2 State of Science and Technology

Autonomous and automated vehicles must have a comprehesive and accurate understanding of their surroundings at all times to ensure safe and reliable operation. This process, known as environment perception, is responsible for detecting, identifying, and tracking objects such as other vehicles, cyclists, pedestrians or other road elements. The quality of the measurement of the state of this objects directly influence the vehicle's ability to make safe and informed driving decisions.

Failures in the perception can lead to hazarous situations, such as collisions or unexpected navigation decisions. This is particularly critical in high-speed scenarios or complex urban environments, where is a high density of high dynamic objects which make more complex the operation in real time.

A robust perception system must provide the following capabilities:

1. **Object detection:** Identify and locate objects in the vehicle's surroundings, such as vehicles, pedestrians, cyclists, or other road elements.
2. **Motion estimation:** Determining the position and velocity of moving objects
3. **Scene understanding:** Classifiying the detected objects and assessing their criticality for the vehicle's navigation
4. **Reliability in edge cases:** Ensuring correct operation in complex and unforeseen scenarios, such as occlusions, adverse weather conditions, or sensor failures.
5. **Real-time operation:** Providing timely and accurate information to the vehicle's control system to make informed driving decisions.

While the perception concept includes different aspects, this work will focus specifically on the detection of moving object in the vehicle's surroundings and the estimation of their 6D motion state (position and velocity) using a deterministic approach. In the following sections, we will provide an overview of the state-of-the-art of the different sensors and algorithms used in the perception systems of autonomous vehicles, focusing on the deterministic image based approaches.

2.1 Perception Techniques in Autonomous Vehicles

The perception systems of autonomous vehicles rely on a combination of different sensors to provide a comprehensive understanding of the ego vehicle's surroundings. In this section we will provide an overview of the most common sensor types used in autonomous vehicles and their advantages and limitations.

2.1.1 LiDAR-Based Perception

LiDAR (Light Detection and Ranging) sensors are increasingly used in the automotive industry for obstacle detection, 3D mapping and lane detection [Van+18]. More and more companies are adopting this type of sensor due to its advantage in rapidly and accurately estimating 3D distances. Examples include BMW [BMW23], Merceces-Benz [Mer25], and Waymo [Way24].

Working principle LiDAR sensors use the concept of time of flight to measure the distance to objects in the vehicle's surroundings. The sensor emits a laser beam and measures the time it takes for the light to reflect back to the sensor. By calculating the time difference, the sensor can determine the distance to the object based on the speed of light [LI20].

Advantages and Limitations LiDAR sensors have the advantage of providing an accurate 3D representation of the vehicle's surroundings. This makes them particularly useful for object detection and localization tasks. However, LiDAR sensors are sensitive to adverse weather conditions, such as rain, snow or fog since the laser beams can be scattered by the particles in the air. Additionally, LiDAR sensors are more expensive than other sensor types, such as cameras or radars, which can limit their adoption in mass-market vehicles, although their prices are steadily decreasing [Van+18].

2.1.2 Radar-Based Perception

Radar (Radio Detection and Ranging) use radio waves to detect objects in the vehicle's surroundings. They are commonly used in the automotive industry for object avoidance.

Working principle Similar to LiDAR sensors, radar sensors emit radio waves and measure the time it takes for the waves to reflect back to the sensor. By calculating the time difference, the sensor can determine the distance to the object based on the speed of light. Radar sensors can also estimate the object's velocity based on the Doppler effect [Leo+08].

Advantages and Limitations Radar sensors are less sensitive to adverse weather conditions than LiDAR sensors, making them more reliable in rain, snow, or fog. [Van+18; RG05]. However, radar sensors have a lower resolution than LiDAR sensors, which can limit their ability to detect small objects or provide detailed information about the object's shape.

2.1.3 Image-Based Perception

Image based perception systems use cameras to capture images of the vehicle's surroundings and are the principal sensor in some companies like Tesla [Maa+20]. These sensors can be used as single cameras for object detection and classification or in pairs to provide stereo vision, which allows for depth estimation and 3D reconstruction of the scene. These type of sensors can also be used for optical flow computation and the ego motion estimation through visual odometry

Monocular vision Monocular vision systems use a single camera to capture images of the vehicle's surroundings. These systems are commonly used for object detection and classification tasks, such as detecting pedestrians, cyclists, or other vehicles [Van+18]. Monocular vision systems are less expensive than stereo vision systems and can be easily integrated into existing vehicles. However, they have limitations in depth perception and 3D reconstruction, which can make it challenging to estimate the distance to objects accurately.

Stereo vision Stereo vision systems use two cameras to capture images of the vehicle's surroundings and use the pinhole camera model [Sze22] to estimate the depth of objects in the scene. By comparing the images from the two cameras, the system can calculate the disparity between corresponding points in the images and use this information to estimate the distance to the objects.

Advantages and Limitations Image-based perception systems have the advantage of providing a rich and detailed information about the environment. Cameras can capture color information, texture, and shape of objects, which can be useful for object detection and classification tasks. In addition, cameras are less expensive than LiDAR sensors and can be easily integrated into existing vehicles. However, image-based perception systems are sensitive to adverse weather conditions, such as rain, snow, or fog, which can affect the quality of the images and limit the system's performance. The algorithms used for object detection and tracking in image-based perception systems can also be computationally intensive, which can affect the system's real-time performance.

In the following table a summary of the advantages and limitations of the different sensor types used in autonomous vehicles is provided:

Sensor type	Advantages	Limitations
LiDAR	Accurate 3D representation	Expensive and sensitive to adverse weather conditions
Radar	Works in bad weather, detects velocity	Low spatial resolution
Cameras	Rich scene details, color information, Cheaper than the other considered sensors	Depth perception is limited, affected by lighting, expensive computation cost

Table 1: Summary of sensor types used in autonomous vehicles

While sensors provide raw data about the environment, perception algorithms are responsible for interpreting this information to detect and track moving objects. These algorithms can be broadly classified into learning-based and deterministic approaches, as discussed in the following sections

2.2 Algorithms for Moving Object Detection

The perception algorithms can also be classified into two main categories: learning-based and deterministic approaches. Learning-based approaches use machine learning techniques, such as deep learning, and are data driven. In the other side, deterministic approaches use mathematical models to estimate the state of the objects in the scene. In the following sections we will provide an overview of the most common algorithms used for moving object detection in autonomous vehicles.

2.2.1 Learning-Based Approaches

Learning-based approaches have gained popularity in recent years. These methods use convolutional neural networks (CNNs), recurrent neural networks (RNNs), and transformers to process the data coming from cameras, LiDARs, and radar systems. In the work done by Wen and Jo [WJ22], they provide a comprehensive review of the different learning-based approaches used in autonomous vehicles, which can be summarized in the following figure:

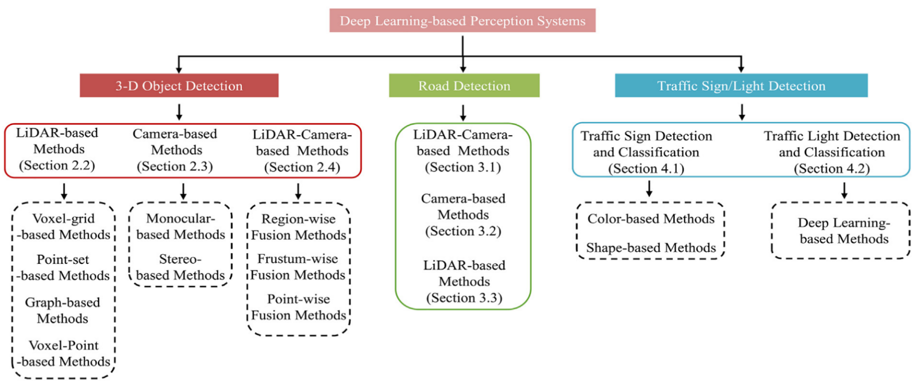

Figure 1: Learning-based perception methods. Source: [WJ22]

Some limitations of learning-based approaches are the need for extensive training data, which can be expensive and time-consuming to collect. These methods can also struggle to generalize in novel and untrained scenarios, which can be problematic in safety-critical situations where the vehicle must make quick decisions based on the perceived environment.

2.2.2 Deterministic Approaches

Deterministic approaches use mathematical models to estimate the state of the objects in the scene.

Optical flow Optical flow is a technique used to estimate the motion of objects in the scene by tracking the movement of pixels between consecutive frames. This method is based on the assumption that the intensity of a pixel remains constant over time, which allows the system to track the movement of objects based on the changes in pixel intensity [Alf+24]. Some of the most common optical flow algorithms are the Lucas-Kanade method [Luc+13] and the Horn-Schunck method [HS81].

6D Vision 6D vision is a deterministic perception framework that combines stereo vision and motion tracking to estimate the 3D position and velocity of objects in the scene. Originally introduced by Franke et all [Fra+05] this method provides a direct motion estimation without requiring object segmentation. The key principles of 6D Vision are:

1. **Stereo disparity estimation:** Calculate the disparity between corresponding points in the stereo images to estimate the depth of objects in the scene.
2. **Optical flow tracking:** Track the movement of feature points in the scene between consecutive frames.
3. **Kalman filtering:** Use Kalman filters to estimate the 6D state vector (position and velocity) of the feature points in the scene.

2.3 State of the Art in the FTM

The Institute of Automotive Technology (FTM) at the Technical University of Munich has conducted previous research in the field of environment perception for autonomous and automated vehicles. This section provides an overview of previous work carried out at the FTM, particularly regarding the EDGAR research vehicle and the development of deterministic perception systems based on 6D vision.

2.4 EDGAR Research Vehicle

The EDGAR (Excellent Driving Garching) research vehicle is a state-of-the-art autonomous driving platform developed at FTM [Kar+24]. It serves as a testbench for autonomous driving research, including perception, planning, control, and sensor fusion. The vehicle is equipped with a multi-sensor setup, comprising:

1. **Stereo and monocular cameras:** For image-based perception and object detection
2. **LiDAR sensors:** For 3D mapping and obstacle detection
3. **Radar sensors:** For object avoidance and velocity estimation
4. **GPS/IMU:** For localization and motion tracking

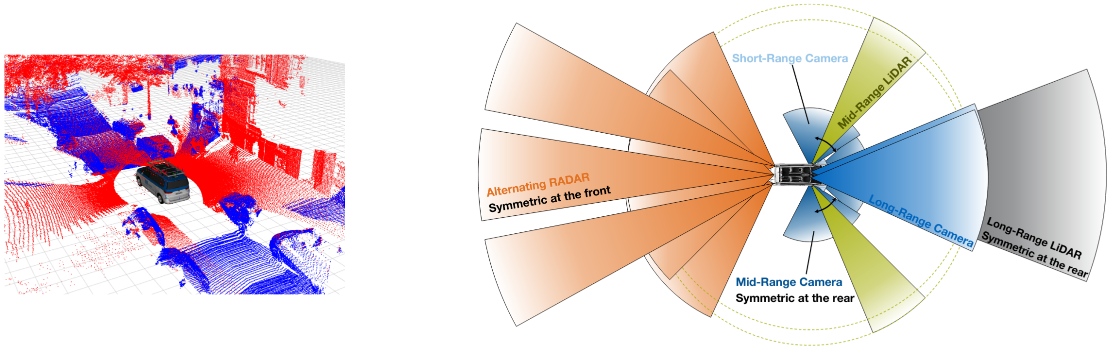

Figure 2: EDGAR Vehicle Sensors

2.5 Previous Work in the FTM

This project builds upon previous work conducted at the FTM in the field of environment perception for automated vehicles with 6D vision. Some of the key contributions include:

1. **Objektdetektion und Bewegungsprädiktion mittels eines Stereokamerasystems beim teleoperierten Fahren** [Bau14]: Bauer in his master's thesis developed a deterministic system for object detection based on the 6D vision of Daimler [Fra+05]. The system was implemented in C++ using the ADTF framework Automotive Data and Time-Triggered Framework and was capable of detecting and tracking objects in the vehicle's surroundings with some real time limitations.

2. **Weiterentwicklung und echtzeitfähige Umsetzung eines Stereo-Vision-basierten Bewegungsprädiktionssystems für die aktive Sicherheit von teleoperierten Fahrzeugen** Some time later in [Rot16] Rotar continued Bauers work by implementing the system in GPU using CUDA to speed up the computationally intensive tasks.

3. **Validation of an Obstacle Detection Approach for Teleoperation of Automated Vehicles:** Finally Pastor [Pas21] started porting the system to ROS1 to facilitate the integration with other modules in the EDGAR vehicle. The system was implemented without CUDA GPU acceleration and was not deployed in the EDGAR vehicle. The system also has some problem with the object detection.

Building upon these previous contributions, this project aims to further improve deterministic perception by addressing key limitations found in prior implementations. In particular, this work will focus on enhancing computational efficiency, real-time feasibility, and system deployment within ROS2. These improvements will be detailed in the next section on objectives.

3 Objective

The goal of this thesis is to develop a deterministic system that can provide a realiable object detection of moving objects in the ego vehicle's surroundings using stereo camera images and the 6D vision concept. For this purpose, the work will be devided in two principal pipelines: the perception pipeline and the criticality assessment pipeline.

3.1 Perception Pipeline

The perception pipeline will be responsible for detecting the moving objects and outputting a list with their position and current velocity. This pipeline will be composed by several submodules as shown in figure 3.

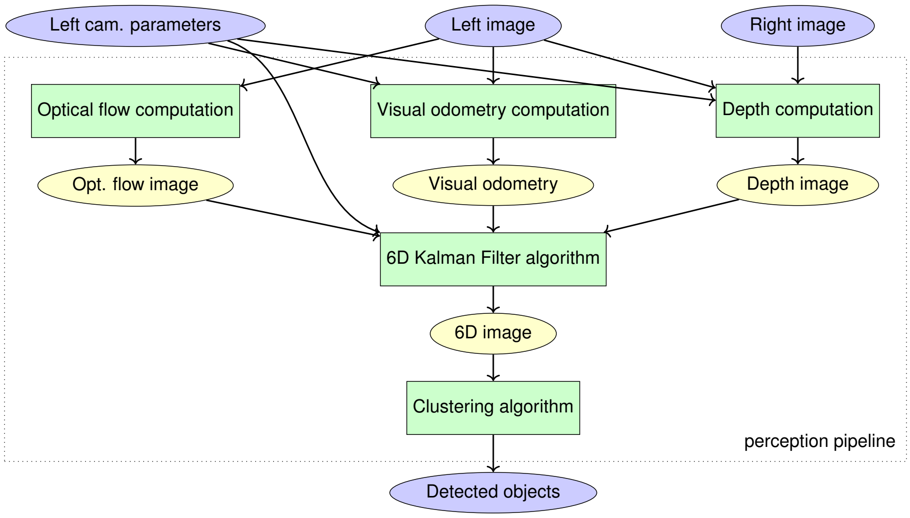

Figure 3: High level perception pipeline

3.1.1 Optical flow computation

This module receives as input the left image data from the stereo camera setup and uses it to compute the dense optical flow field.

3.1.2 Visual odometry computation

To compute the motion field of the environment, its necessary to know how the ego vehicle is moving with respect to the environment.

Odometry is the use of data from motion sensors to estimate changes in position over time. It is commonly used in robotics and autonomous vehicles to track the movement of the vehicle.

Visual odometry, on the other hand, is a specific type of odometry that uses visual data (images or video) to estimate the motion of the vehicle. By analyzing the changes in the visual data over time, the system can infer the vehicle's movement and position relative to its environment. In order to build the pipeline independent of the hardware used, a visual odometry module will be implemented to estimate the ego vehicle's motion.

Visual odometry can be divided into two categories: monocular and stereo visual odometry.

- **Monocular visual odometry:** Uses a single camera to estimate the motion of the vehicle [He+20].

- **Stereo visual odometry:** Uses a stereo camera to estimate the motion of the vehicle [LB20], including also the depth information.

Since the system is intended to be used with stereo cameras, and the depth information is already computed for the 6D vision algorithm, the stereo visual odometry will be implemented.

3.1.3 Depth computation

Even through the EDGAR vehicle is equipped with a stereo camera as will be explained in section 3.3, to ensure the system is robust and can be tested with other camera configurations or datasets, a depth computation module will be implemented to compute the depth map from the stereo images. This module will take as input the left and right images, as well as the camera calibration parameters (focal distance, principal point and baseline [Sze22]) and output the depth map of the scene.

3.1.4 6D Kalman Filter algorithm

This module will be responsible for estimating the 6D motion of the objects and will serve as the backbone of the perception pipeline. It will take the dense optical flow field, the depth image and the ego vehicle's motion as input and output the 6D motion of the objects. The output will be an image with 6 channels to represent the 6D state of each tracked point in the scene.

3.1.5 Clustering algorithm

To group the 6D motion vectors into objects, a clustering algorithm [JMF99] will be implemented. The clustering algorithm will take the 6D motion image as input and output a list of objects with their 6D motion vectors. For this purpose, the DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm will be used [Den20]. This clustering algorithm is suitable for this task since it can handle noise and outliers and does not require the number of clusters to be specified like in other clustering algorithms like k-means [ASI20].

3.2 Criticality Assessment Pipeline

The other pipeline (figure 4) will be responsible for assessing the criticality of the detected objects. This pipeline will be composed by the criticality algorithm that will compute metrics like the time to collision (TTC) [Wes+23], based on the object list outputted by the perception pipeline and the ego vehicle's trajectory.

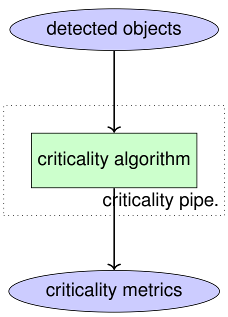

Figure 4: High level criticality pipeline

3.3 Development and real time implementation

For the development of the system, the stereo camera setup of front of the EDGAR vehicle will be used. This consist in two Framos D455e stereo cameras, two long range Basler cameras in stereo configuration and three medium range Basler cameras for object detection. These can be seen in the following figure:

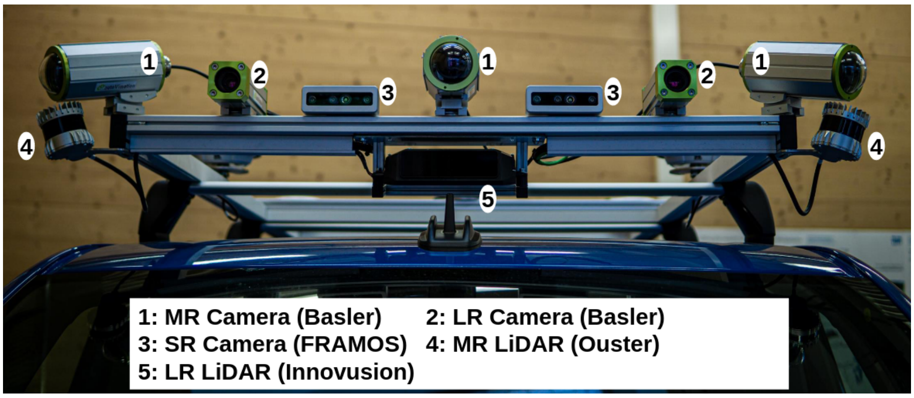

Figure 5: EDGAR vehicle camera setup

From this setup, only the Framos and the long range basler cameras will be considered since they are the ones that provide the stereo images.

Regarding the software implementation, the system will be developed using the Robot Operating System (ROS) framework [ros25] due to its modularity and easy integration with the EDGAR vehicle's existing software. This framework supports two primary programming languages: C++ and Python. Since C++ offers higher performance, it will be used to implement the system.

The EDGAR vehicle employs an architecture based on Docker images for each service. Docker is a platform that automates the deployment of applications within lightweight, portable containers. These containers bundle an application together with all its dependencies, libraries, and configuration files, ensuring consistent behavior across different environments. Consequently, the system must also be containerized using Docker for deployment in the vehicle.

Regarding the real-time implementation, extensive image processing makes a multi-threaded system essential. One of the most powerful tools for this purpose is NVIDIA's CUDA [NVI07], an API that allows the GPU to be used for general-purpose computing rather than just image rendering. This capability can be tapped either by writing custom CUDA kernels [Mah19] or by using libraries like OpenCV [Cer20] that include GPU support. Consequently, GPU acceleration will be considered for all computationally intensive tasks such as optical flow, clustering, and depth computation.

3.4 Evaluation

For the evaluation, two aspects will be assessed separately: the individual components of the perception pipeline and the integrated system.

To evaluate the optical flow, depth computation, and visual odometry modules, the corresponding datasets from the KITTI (Karlsruhe Institute of Technology and Toyota Technological Institute) Vision Benchmark Suite [Gei+13] will be used. These datasets provide a diverse range of real-world driving scenarios—including urban and highway scenes—along with ground truth data for optical flow [MHG18], depth [Uhr+17], and visual odometry [GLU12].

To assess the integrated system, the goal is to compare its performance with the state-of-the-art learning-based perception systems currently deployed in the EDGAR vehicle. For this purpose, the developed system will be installed on the EDGAR platform and tested in various environments, such as urban settings and parking lots. In each scenario, the criticality metrics produced by the system will be compared with those generated by the existing perception solutions.

In addition to evaluating output data, another key metric will be the system's computational time. Specifically, the frame rate of each module will be measured, and the overall frame rate of the system will be calculated. This is crucial because the system is intended for real-time operations, where low computational overhead is n ecessary to maintain the required frame rate. These measurements will then be compared with those of the current perception systems used in the EDGAR vehicle.

3.5 Expected Results

To summarize the anticipated outcomes of this thesis and how they will advance the department's ongoing work, the following results are expected:

- Development of a deterministic, 6D vision-based perception system for moving object detection on autonomous vehicles.
- Implementation in C++ using the ROS framework, with containerization via Docker for streamlined deployment on the EDGAR vehicle.
- Integration of GPU acceleration through CUDA, leveraging the work of [Pas21] to improve performance in computationally intensive tasks.
- Capability to estimate criticality metrics, such as time to collision (TTC), for identified objects.
- Evaluation of individual modules using the KITTI dataset and validation of the integrated system on the EDGAR vehicle.
- Enhancement of the existing clustering method by employing the DBSCAN algorithm for more robust object detection.

4 Workplan

In the following sections, the work plan for the thesis is presented. The work plan is divided into five main work packages (WPs), each with a set of tasks and deliverables. This are:

- **WP1: Literature research and planning:** This WP comprises all the tasks that are necessary to understand the state-of-the-art of the frame of the project and learn everything that is necessary to develop the necessary algorithms. It also includes the planning phase of the project.
- **WP2: Development of the perception pipeline:** This WP is dedicated to the development, fine-tuning and testing of the perception pipeline, which is responsible for detecting the moving objects in the scene.
- **WP3: Development of the criticality assessment pipeline:** This WP is dedicated to the development, fine-tuning and testing of the criticality assessment pipeline, which is responsible for assessing the criticality of the detected objects.
- **WP4: Evaluation and discussion of the result:** This WP is dedicated to the deployment of the developed system on the EDGAR vehicle and the evaluation of the results. The results will be compared against the state-of-the-art learning-based perception systems currently used in the vehicle. For this will be necessary to collect data from the vehicle and analyze it.
- **WP5: Materthesis project management:** This WP is dedicated to the management of the masterthesis project. It includes the writing of the thesis, the preparation of the presentation and the final defense of the thesis.

In the figure 6 the time schedule of the project is presented as an Gantt Diagram.

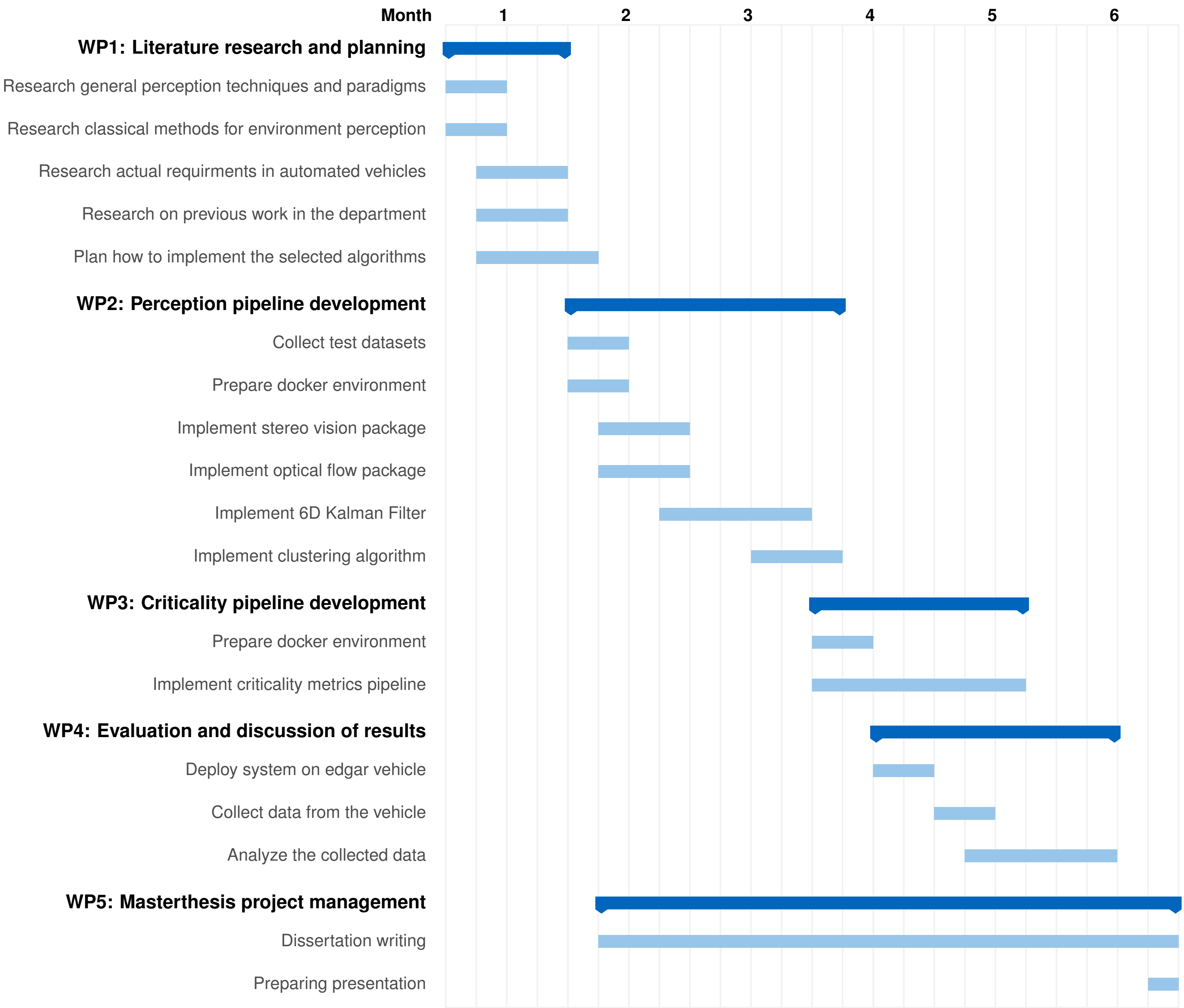

Figure 6: Time schedule.

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